

Danish Syed

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EDUCATION	University of Michigan <i>Masters, Electrical & Computer Engineering - Computer Vision</i> • GPA: 3.75/4.00 Indian Institute Of Technology (ISM) <i>B.Tech, Electronics & Communication Engineering</i> • GPA: 8.75/10	Ann Arbor, USA Sep 19 - Present Dhanbad, India Jul 13 - May 17
RESEARCH EXPERIENCE	Johnson AI Lab, University of Michigan <i>Graduate Student Researcher Advisor: Prof. Justin Johnson</i> • Working on generatively designing complex architectural spaces conditioned on physical properties like sound quality for operas and crowd flow for airports • Designed 3D correlates of 2D interpretability methods like feature inversion to understand the learned features of pointcloud & mesh based deep neural networks Taubman College of Architecture, University of Michigan <i>Research/Teaching Assistant Advisor: Prof. Matias Campo</i> • Worked with architecture grad students (<i>as part of ARCH662</i>) to explore whether the current state of AI can have a novel sensibility of human creativity at large • Implemented various style transfer methods (<i>GAN and VGG based</i>) to empirically explore the hypotheses devised by the students [https://arch-ai.org]	Ann Arbor, USA May 20 - Present Ann Arbor, USA Jan 20 - Sep 20
PUBLICATIONS	DeepGamble: Towards unlocking real-time player intelligence using multi-layer instance segmentation and attribute detection Danish Syed* , Naman Gandhi*, Arushi Arora* & Nilesh Kadam <i>ICMLA 2020 (Full Paper)</i> [arXiv]	
TECHNICAL SKILLS	• Languages: Python, C/C++, SQL, MATLAB, Julia • Frameworks: PyTorch, TensorFlow, CUDA, TensorRT • Tools: Docker, AWS, Slurm, Git, JIRA, Jupyter, L ^A T _E X	
PROFESSIONAL EXPERIENCE	Intvo Inc. <i>Computer Vision Engineer Intern</i> Built an ML pipeline to understand pedestrian behavior in the context of autonomous cars • Designed data gathering strategies to supplement open-source pedestrian behavior datasets • Implemented a novel deep learning model to predict pedestrian motion, awareness, etc. • Used PyTorch and TensorRT libraries to deploy & test the model in the city of Ann Arbor ZS Associates <i>Data Scientist</i> Casino & Gaming Industry Clients • Designed a video recognition system (<i>DeepGamble</i>) to create a digital understanding of blackjack players, which proposed over \$100M of savings annually	Ann Arbor, USA Jun 20 - Aug 20 Pune, India Jul 17 - May 19

* Indicates equal contribution

- Performed Monte-Carlo simulations to understand the customer journey and luck induced volatility in their gameplay behavior, to tailor the promotion techniques

Pharmaceutical Industry Clients

- Implemented an LSTM autoencoder to analyze the real-time sensor data and predict anomalies in their batch and suggested key-drivers for non-conformance
- Identified Key Opinion Leaders in the health-care domain from Twitter data by leveraging page-rank algorithm

RELEVANT PROJECTS

SPPFNet (*EECS 542: Advance Computer Vision Course Project*)

Attempted to build a particle filter based pose estimator where each particle learns latent embedding to infer pose, object likelihood, and re-sampling objective iteratively

SICGAN (*EECS 504: Computer Vision Course Project*)

Implemented an end-to-end GAN framework for generating 3D meshes from a single image. We were able to get qualitatively better 3D reconstructions as compared to the baseline

MedLens (*ZS Quest Hackathon Winners*)

Used a Bi-Directional Attention Flow network to develop a web app for health-care researchers, which answers factual questions based on uploaded documents or [PubMed](#)

Kaggle Projects (*Silver/Bronze Medals*)

- Built a ConvNet to identify atmospheric conditions & land type from satellite images
- Experimented with U-Nets to count different types of sea lions in aerial photographs
- Built an ensemble of LSTM and XGBoost models to predict similar questions on Quora

TEACHING

GSI, EECS 442: Computer Vision (CV), University of Michigan
GSI, EECS 598: Deep Learning for CV, University of Michigan
TA/RA, ARCH 662: Architecture & AI, University of Michigan
Instructor, Architecture & AI Workshop, DigitalFutures

Winter 21
 Fall 20
 Winter 20
 Winter 20

SALIENT COURSES

- **UMICH**: Computer Vision, Deep Learning for Computer Vision, Mobile Robotics, Probability & Random Processes, Matrix Methods, Applied GPU Programming
- **IIT(ISM)**: Machine Learning, Digital Image Processing, Computer Programming, Data Structures, Database Management System, Numerical & Statistical Methods

EXTRA CURRICULAR

Director, TEDx, IIT(ISM) Dhanbad
Club Head, Robotics Club, IIT(ISM) Dhanbad
Tutor, Kartavya (a student run NGO), IIT(ISM) Dhanbad